

Project Title
(Hospital Bed / OR Scheduling)

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Hospital Bed and Operating Room Scheduling is a challenging task because hospitals have limited beds, operating rooms, staff, and time slots, while patients may have different priorities and treatment requirements. This project focuses on developing a scheduling engine that helps hospitals assign patients and medical procedures to suitable beds, operating rooms, and available time slots. The system considers factors such as procedure duration, patient priority, staff availability, room capacity, and treatment deadlines while creating the schedule. It aims to reduce unnecessary waiting time, room idle time, and scheduling conflicts, while making better use of available hospital resources. The system also provides quality reports showing important measures such as room utilization, patient waiting time, and overall scheduling efficiency. By automating the scheduling process, the project can help hospital staff manage resources more effectively and make faster scheduling decisions. Overall, the system provides a practical approach to improving hospital operations and patient flow.

Signature of the Guide